



Detection of ADHD cases using CNN and classical classifiers of raw EEG

Behrad TaghiBeyglou^{a,*}, Ashkan Shahbazi^b, Fatemeh Bagheri^c, Sina Akbarian^d, Mehran Jahed^e

^a Institute of Biomedical Engineering, University of Toronto, Toronto, ON, Canada

^b Biomedical Engineering Department, Amirkabir University of Technology, Tehran, Iran

^c Medical Biophysics Department, University of Toronto, Toronto, ON, Canada

^d Vector Institute for Artificial Intelligence, Toronto, ON, Canada

^e School of Electrical Engineering, Sharif University of Technology, Tehran, Iran

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ABSTRACT

Purpose: This study proposes a novel convolutional neural network (CNN) structure in conjunction with classical machine learning models, utilizing the raw electroencephalography (EEG) signal as the input to diagnose attention deficit hyperactivity disorder (ADHD) in children. The proposed EEG-based approach does not require transformation or artifact rejection techniques.

Methods: In the first step, the suggested method uses raw EEG to train a CNN to diagnose ADHD. Then, the feature maps from different layers of the trained CNN are extracted and used to train some classical classifiers such as support vector machine (SVM), logistic regression (LR), random forest (RF), etc. This study benefits from an extended version of a dataset acquired from 61 participants diagnosed with ADHD and 60 individuals in control group, age 7 through 12 years old.

Results: The initial CNN structure (without further use of feature maps) achieved an accuracy of $86.33 \pm 2.64\%$ in 5-fold cross-validation scheme on training set, which is superior to results reported in previous studies. However, in order to increase the efficacy of the classifiers we used various feature representations across different CNN layers and after a rigorous evaluation of candidate classifiers, logistic regression provided an accuracy of $91.16 \pm 0.03\%$ in training epochs using 5-fold cross-validation scheme and 95.83% in ADHD identification in unseen epochs, were achieved. Also, other metrics such as precision, sensitivity, F1-score and receiver of operating characteristic (ROC) were presented for better comparison of different hybrid methods.

Conclusion: The suggested method for detection of ADHD in children shows high performance in different metrics such as accuracy, sensitivity, and specificity, which is superior to previously reported results.

1. Introduction

Attention deficit hyperactivity disorder (ADHD) is a disease group related to various dysfunctions [1]. There are two different words in the name of this disorder, and each of them is responsible for a specific symptom of ADHD. The first group of patients with attention deficit can rarely focus on a unique topic, and anything surrounding them in their environment can distract them quickly; however, the second group is related to hyperactivity symptoms, where they can tolerate a static situation and intend so much to alter their posture and position, in some cases, they may shout or keep silence suddenly in a regular discussion [2–4]. All these conditions can happen for children in different stages of growth [5–7].

Based on the Centers for Disease Control and Prevention (CDC) statistics in 2016, over 6.1 million children in the age group of 4–17 years have been diagnosed with ADHD [8]. Overall, patients can

be categorized into three different stages of ages, the children in a range of 2–5 years are 388 thousand, the children in the age of 6–11 are 4 million, and the rest are in the range of 12–17 years old. According to the National Institute of Mental Health (NIMH), around 70% of children with ADHD continue to have some hyperactivity and impulsivity during their adolescence and adulthood [9]. Long-term follow-up studies from childhood to adulthood revealed that children with ADHD, compared with those without ADHD, were more involved in psychosocial, educational, and neuropsychological functioning and had higher risks for antisocial disorders, major depression, and anxiety disorders as adults [10–12].

With the development of medical technologies and the application of various analytical tools, research to extract the brain's functionality during ADHD, its electrical patterns and connectivity are on the rise [13]. In recent years, many researchers have focused their

* Corresponding author.

E-mail address: behrad.taghibeyglou@mail.utoronto.ca (B. TaghiBeyglou).

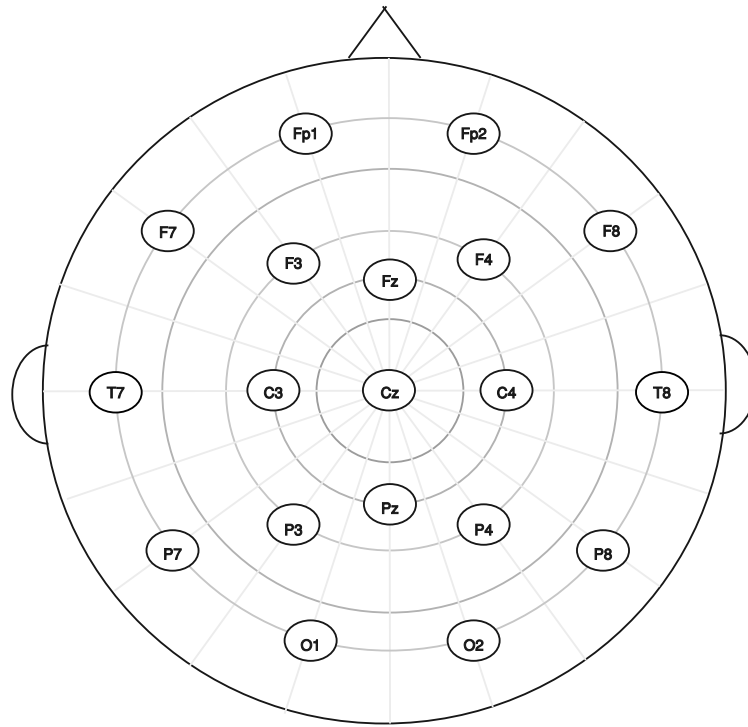


Fig. 1. Channel location of recorded signals in dataset used in this research [16].

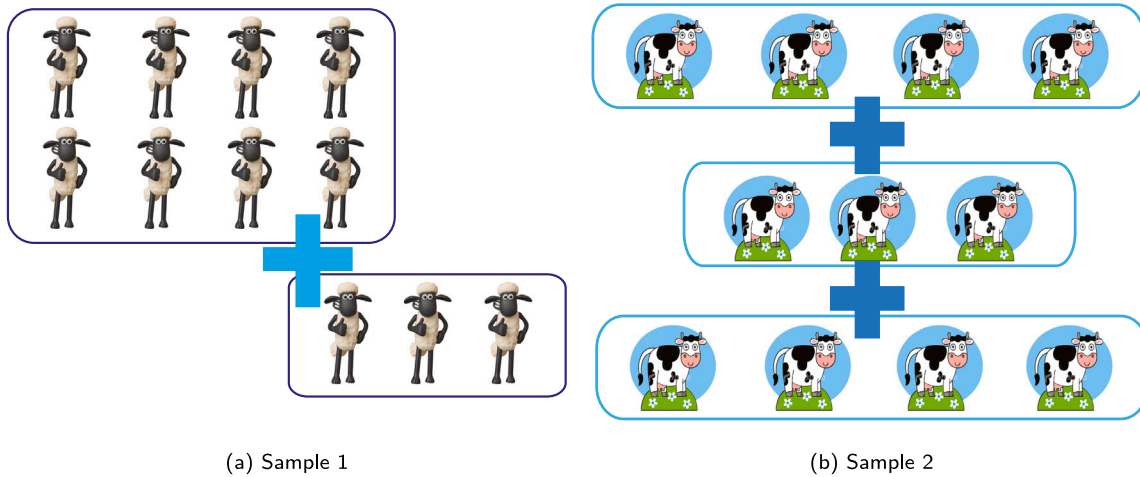


Fig. 2. Two sample tasks of dataset [16].

attention on using different neuroimaging modalities and biomedical signal processing techniques to determine the behavior and origin of ADHD as early as possible. Although finding out the mechanism and causality seems slow, the recognizance techniques are improving fast, but it should be mentioned that some clinical assessments such as The Diagnostic and Statistical Manual of Mental Disorders (DSM) [14], and International Classification of Diseases (ICD) [15] were introduced. Nevertheless, the diagnosis procedure is time-consuming and subjective; therefore, the results can easily vary from one case to another. That is why introducing a biomarker that uses individuals' characteristics and performs fast and accurately can play a beneficial role.

1.1. Background

The EEG modality has high temporal resolution and can provide high spatial resolution in high density (electrodes) recording. The EEG

can also be acquired during the movement and routine daily life, even from the children with hyper activities [17–19]. These advantages have led many researchers to benefit from EEG signals in their ADHD-related projects.

1.2. Literature review

Before applying deep neural network (DNN) structures in ADHD detection, many researchers have been extracting features manually with notable results [5,20–28]. Nowadays, with huge advancements in DNNs in the broad application of biomedical signal processing, it is noteworthy to apply these kinds of algorithms in ADHD detection to extract features and classify based on observations automatically. For example, some researchers have used connectivity in conjunction with neural networks to diagnose ADHD from EEG signals [29,30]. In other group of previous research, some researchers have been focusing on time–frequency related characteristics of EEG signals to build

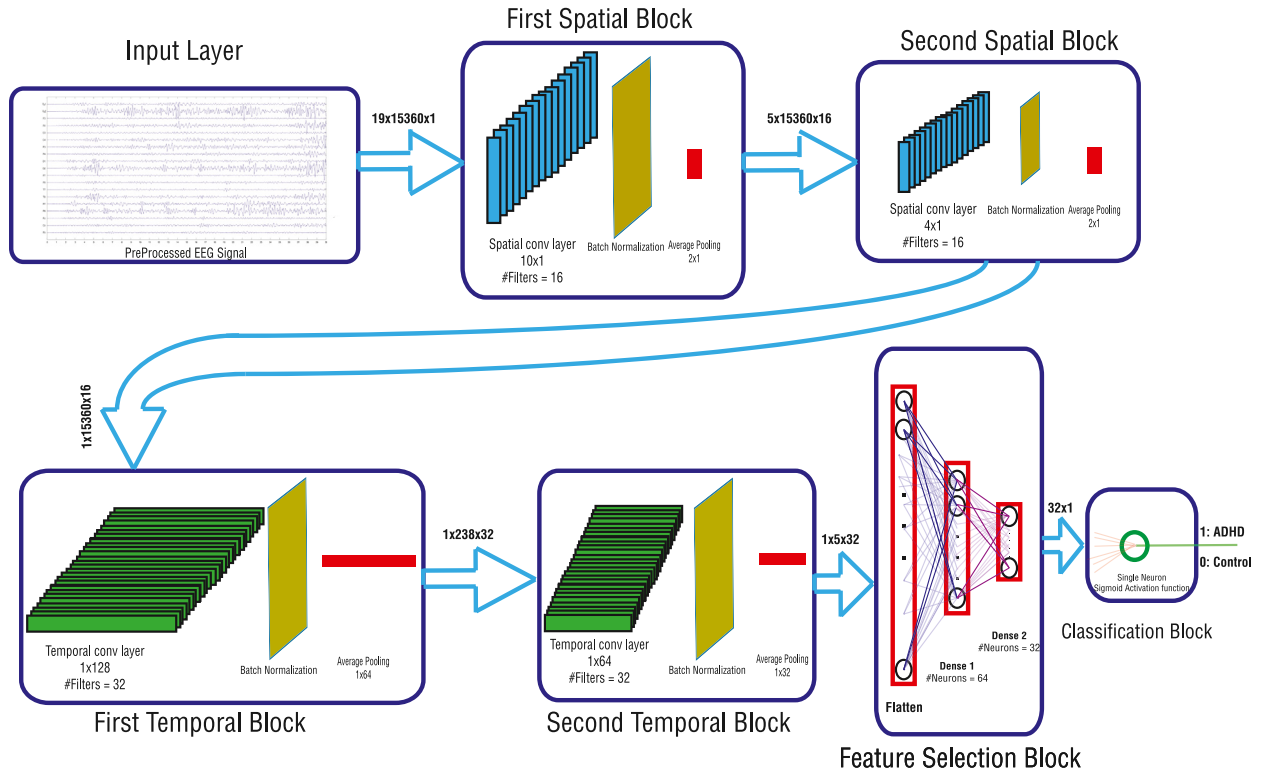


Fig. 3. This figure shows the proposed CNN structure. The filter size is figured proportional to the real values. The blue, green, yellow, and red colors are related to 1D spatial constitutional, 1D temporal convolution, batch normalization, and average pooling layers, respectively.

and develop neural networks [31–33]. However, no previous research have tried to utilize raw EEG in developing DNN to classify ADHD in children.

1.3. Contribution

This research presents a new framework which is benefited from the CNN structures in order to extract discriminative features and accordingly distinguish ADHD and from Control individuals. Notably, we only feed our neural network with raw EEG signals and consequently there is no need to utilize any specific transformations such as wavelet, time–frequency, or spatio-temporal filtering methods, and our proposed framework does not need specialized techniques such as re-referencing, artifact rejection, etc. Also, we use the classical classifiers to benefit from the formulations in solving the non-linearity of the produced features at different layers of the CNN.

In the following sections, initially we will discuss the data description, the architecture of the proposed CNN, proposed classical classifiers used in this research, and the overall framework utilized for detecting ADHD in children. Section 3 shows our results using various performance metrics through three different feature mapping techniques, thus comparing our result with selected and newest literature in deep learning and manual feature selection techniques for ADHD detection in children. Finally, in Section 4, the findings containing our proposed method’s advantages and limitations are discussed.

2. Materials and methods

2.1. Data description

We used the dataset for evaluating our framework, which recorded 61 ADHD and 60 healthy children as the participants. The children were aged between 7 to 12 years old and inclusive of both girls and boys. The ADHD disordered children were diagnosed by a psychiatrist

to DSM-IV criteria and have taken Ritalin for up to 6 months [16]. None of the control group children had a history of psychiatric disorders, epilepsy, or any report of high-risk behaviors [16]. EEG signals were acquisition based on 10–20 standard from 19 electrodes (Fz, Cz, Pz, C3, T3, C4, T4, Fp1, Fp2, F3, F4, F7, F8, P3, P4, T5, T6, O1, O2) which are shown in Fig. 1 at 128 Hz sampling frequency. The earlobe electrodes named A2 and A2 were used as the reference channels. We used a more refined version of the mentioned dataset, which was first used in EEG competition by the National Brain Mapping Laboratory (NBML).

As one of the symptoms in ADHD children is attention (primarily visual attention), the recording protocol was set about visual attention tasks. In the mentioned task, a set of cartoon characters in pictures are shown to the children and then asked to count the characters and perform the illustrated character’s computation task. The number of characters in each image was randomly selected between 5 and 16, and the size of the pictures was large enough to be easily visible and countable by children [16]. In Fig. 2, two sample tasks are shown. Also, for recording continuous EEG, the pictures and the tasks are demonstrated uninterrupted and immediately after each other to measure the child’s response at each epoch. Thus, the duration of EEG recording throughout this cognitive visual task depended on the child’s performance (i.e., response speed) [16].

However, we used the modified version of the dataset developed for the national EEG processing competition and held by the NBML [34]. In that version, the dataset was resampled to a 512 Hz sampling rate, and the dataset of 30 s of the original recording containing computational tasks was extracted. Due to the primary dataset’s actual response time, each subject had 4–7 epochs with 30 s duration. The training set contains 196 ADHD and 132 control epochs from one group of subjects, while the validation set contains 37 ADHD, 35 healthy epochs from the other group of subjects. The training and validation epochs had been extracted from different subjects without any intersection. That is why our analysis and results on the validation set lead to an unbiased conclusion.

Table 1
Layer properties of the proposed CNN structure.

Block name	Layer name	Number of filters	Filter size	Output shape	Trainable parameters
Input	Input layer	–	–	$19 \times 15360 \times 1$	0
First spatial	Convolution	16	10×1	$10 \times 15360 \times 16$	176
	Batch normalization	–	–	$10 \times 15360 \times 16$	64
	Average pooling	1	2×1	$5 \times 15360 \times 16$	0
Second spatial	Convolution	16	4×1	$2 \times 15360 \times 16$	1040
	Batch normalization	–	–	$2 \times 15360 \times 16$	64
	Average pooling	1	2×1	$1 \times 15360 \times 16$	0
First temporal	Convolution	32	1×128	$1 \times 15233 \times 32$	65568
	Batch normalization	–	–	$1 \times 15233 \times 32$	128
	Average pooling	1	1×64	$1 \times 238 \times 32$	0
Second temporal	Convolution	32	1×64	$1 \times 175 \times 32$	65568
	Batch normalization	–	–	$1 \times 175 \times 32$	128
	Average pooling	1	1×32	$1 \times 5 \times 32$	0
Feature selection	Flatten	–	–	160×1	0
	Dense 1	64	1	64×1	10304
	Dense 2	32	1	32×1	2080
Classification	Dense 3	1	1	1	33

2.2. CNN

CNNs are the types of neural networks to extract many detailed points from the images. This research wants to feed the network with raw EEG signals as the network's input. Our methodology considers the CNN as the feature extractor from spatial (channel) and temporal (time) information. The spatial information helps the network understand the relationship between the electrodes and the time-series values simultaneously; however, the temporal information helps the network extract the relationship between the time series samples on the same channel. Our proposed network's novelty uses a double layer to extract two-stage of pure information in both spatial and temporal spaces. The layers and blocks we had used in this research can be overview below. Also, the overall scheme of our network is shown in Fig. 3.

- As mentioned earlier, we are feeding our network with the pre-processed EEG signals, which are at the same dimension as the raw EEG signals, so the input layer must be in the shape of Channels (C) $\times$ Time-samples (T). As we have the signals in the form of 19 channels and 30 s epoch, which is equal to $30 \times 512 = 15360$ samples, the input shape of the network is a tensor in size of $(19 \times 15360 \times 1)$ for each sample.
- After feeding the network, it is time to find the spatial relationship between the channels. For this purpose, at the 1st spatial block, we used 16 filters with the size of (10×1) for weighted averaging between different channels to gain a more pure EEG signal, making the overall network stronger recognizing ADHD from non-ADHD epochs. In this block, we also used a batch normalization layer to increase the regularization of the network and make the neurons' weights and filters less biased and increase the robustness of the overall network (The algorithm is fully shown [35]). At the 1st spatial block, the final layer is the Average pooling layer with the size of (2×1) , which means that it computes the average of two-channel samples and returns them output the block.
- The next block is the 2nd spatial block, which exerts the previous block signals, and then by using 16 filters in size of (4×1) computes another weighted averaging between the channels after that Batch Normalization and Average Pooling layers with the specifications of the previous block are utilized again.
- Now, we had a signal in the tensor with the size of $1 \times 15360 \times 16$, and it tells us that we should apply temporal filtering to elicit the time-series at this stage relation. For that purpose, we used

32 filters with a size of (1×128) to ensemble averaging between the samples. The kernel size 128 is selected with the $F_s/4$ and related to the sampling frequency characteristic. After that again, Batch Normalization and Average pooling with the size of (1×64) . It should be mentioned that the size of the pooling window is picked up with half of the size of temporal kernels. We called this overall block containing one temporal filtering layer, one Batch normalization layer, and one Average pooling layer, first temporal block.

- After the 1st temporal block, it is worthy of being more depth into the temporal information and exerting another temporal filtering layer with 32 filters and the size of (1×64) which is selected based on $F_s/8$. After that again, we used the Batch Normalization and one Average Pooling layer with a size of (1×32) and called these three layers as the 2nd temporal block.
- Up to here, we could extract precise information from both spatial and temporal filtering, but extracted features should be best selected for increasing the performance of the whole network. For doing that, we need to Flatten the information after all four previous blocks to prepare to participate in feature selection in the last layers.
- For this feature selection paradigm, we used two Dense (Fully connected) layers with a size of 64 and 32, respectively. We call these two layers to feature selection block because of their functionality.
- At the final block, it is needed to have a single neuron with a sigmoid activation function to compute the probability of each class (0 for Control, 1 for ADHD) and building up the loss function. This block can be named the classification block.

As different layers in CNN are acting not uniquely. It is noteworthy to mention the sequencing and also input and output shapes of the layers. Also, it should be noted that the activation function of all layers except the last dense layer (which is sigmoid) is the Rectified Linear Unit (ReLU). For that purpose, the characteristic of different layers of the network is mentioned in Table 1.

2.3. Classical classifiers

In this research, different classifiers were used to be coupled with a CNN structure. The list and the main parameters of each one are mentioned below (All the implementations were done on Intel Core 17-7700HQ, NVIDIA GeForce GTX 1050Ti, 16 GB DDR4 RAM, Ubuntu 16.04 LTS, and Python 3.5.6):

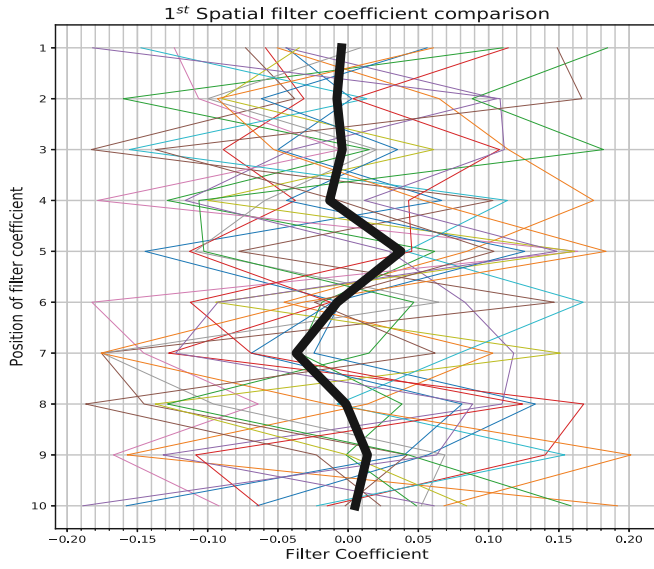


Fig. 4. First spatial block filters coefficients (each color shows one filter), and the average of these filters are shown in thick black line.

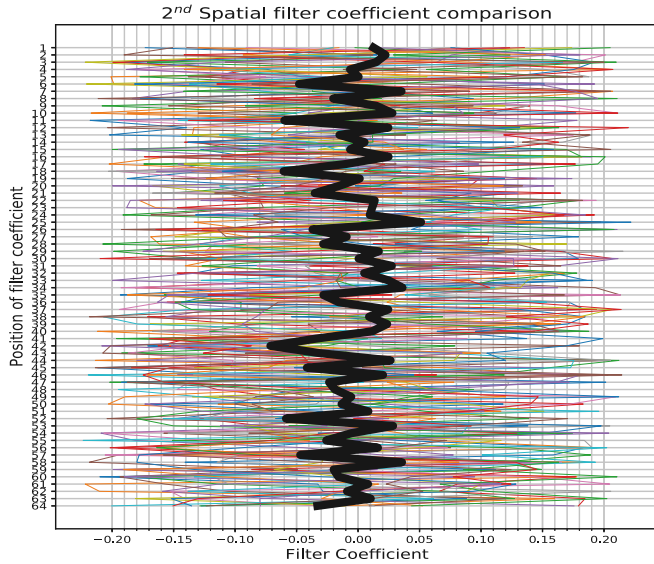


Fig. 5. Second spatial block filters coefficients (each color shows one filter), and the average of these filters are shown in thick black.

- Nonlinear (kernelized) Support Vector Machine (NLSVM) with Radial Basis Function (RBF) kernel with optimizing the scale and gamma as the hyper-parameters.
- Logistic Regression (LR) with L2-norm penalties.
- Random Forest (RF) with 100 number of trees.
- Gaussian Naive Bayes (GNB).
- K-Nearest Neighbor (KNN) with $K = 5$ neighbors.

2.4. Hybrid proposed method

Since the CNN's last layer has a dense layer, this single neuron will function as a classifier, returning the likelihood of belonging to the ADHD category. From the last layers of the proposed system, we can extract pure features that have passed different spatial and temporal blocks for better performance and use them for training the classical classifiers listed in the previous section. It seems this hybrid method benefit from these advantages:

Table 2

CNN classification results over 10 repeated training times.

Metric (range)	Value (mean $\pm$ STD)
Accuracy (0%-100%)	86.33 $\pm$ 2.64
Precision _{ADHD} (0-1)	0.89 $\pm$ 0.04
Precision _{Control} (0-1)	0.85 $\pm$ 0.07
Recall _{ADHD} (0-1)	0.89 $\pm$ 0.07
Recall _{Control} (0-1)	0.82 $\pm$ 0.09
F1-score _{ADHD} (0-1)	0.88 $\pm$ 0.03
F1-score _{Control} (0-1)	0.83 $\pm$ 0.03
AUC (0-1)	0.86 $\pm$ 0.03

- The features are extracted entirely automatically. The overall scheme focuses on extracting as many separable features as possible from raw signals, so there is no expert comment during the feature extraction procedure.
- Since one single neuron is very shallow for discrimination of different groups in cases of non-linear features in complex dimensions, the overall structure will replace the dense layer classification. As a result, integrating non-linear classifiers improves the separability of ADHD and general classes, as well as the sensitivity and specificity, which are important metrics to consider when assessing a disease detection system.

This research finally used extracted features (feature maps) from each dense layers' output. This leads to extract features after the Flatten, Dense 1, and Dense 2 layers with dimensions 160, 64, and 32. After extracting the features, the classical classifiers are trained to perform the discrimination between ADHD and Control.

3. Results

In this section, we first introduce the set-up parameters for training our CNN model, and then we exploit the best scheme for training the overall CNN. Finally, we check out the hybrid possibility of combining the CNN and classical classifiers to achieve better performance.

3.1. Training the CNN

For training the overall CNN, we devoted 70% of our training data for training the CNN and the rest for validating the training procedure. The training and validation scheme ratio was kept consistent while different segments are data were selected because of the existence of the shuffling of the data at each epoch of training. We used the Binary Cross Entropy (BCE) as the loss function of the training. Also, Adam optimizer [36] was used to find the networks' weights and the filters to minimize the loss function with the backpropagation technique by setting up the learning rate 0.0001. The number of total epochs was also set to 100, and each one dependent on the batch size mentioned above; there would be different iterations for training the model inside the epoch. The validation set's loss and accuracy are computed at the end of the epoch and in the final iteration to measure the training's performance on both the training and validation sets.

As the most prominent challenge during neural network training is overfitting to the training data and inappropriate results on the test data, we examined two different approaches for training our deep neural network structure to avoid overfitting; (1) Early stopping with 30 epochs patience (2) Best validation accuracy model. After running the algorithm more than ten times in each strategy, the best validation accuracy model had been selected for further analysis.

After training the neural network and finding the weights and gains of different filters, we should assess the relationship between the filters and the physiological concepts. For that purpose, we extract the filter coefficients and visualize them in the following figures. In Fig. 4 the coefficients of the first spatial filters show that the most impacting

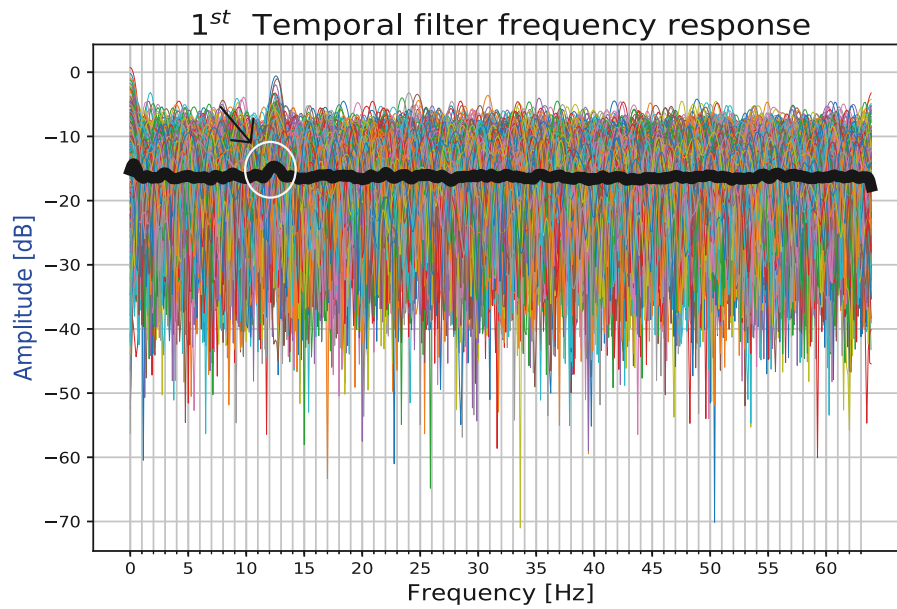


Fig. 6. Frequency response (in dB) of the first temporal block filters in respect to the Hz. The circled section demonstrates that attention of the filters is on the alpha band of EEG signals.

Table 3

Results of hybrid classification scheme comparison using Flatten layer feature maps (160-dimension) in 5-fold cross validation.

Metric (Range)	NLSVM	LR	RF	GNB	KNN	CNN
Accuracy (0%–100%)	85.67 $\pm$ 3.15	87.82 $\pm$ 0.03	85.96 $\pm$ 3.87	82.92 $\pm$ 5.87	94.52 $\pm$ 3.27	86.33 $\pm$ 2.64
Precision _{ADHD} (0–1)	0.9 $\pm$ 0.05	0.93 $\pm$ 0.02	0.9 $\pm$ 0.02	0.84 $\pm$ 0.06	0.96 $\pm$ 0.05	0.89 $\pm$ 0.04
Precision _{Control} (0–1)	0.79 $\pm$ 0.03	0.81 $\pm$ 0.08	0.81 $\pm$ 0.07	0.81 $\pm$ 0.07	0.93 $\pm$ 0.03	0.85 $\pm$ 0.07
Recall _{ADHD} (0–1)	0.85 $\pm$ 0.02	0.87 $\pm$ 0.03	0.86 $\pm$ 0.05	0.88 $\pm$ 0.05	0.95 $\pm$ 0.02	0.89 $\pm$ 0.07
Recall _{Control} (0–1)	0.86 $\pm$ 0.06	0.9 $\pm$ 0.05	0.85 $\pm$ 0.05	0.76 $\pm$ 0.1	0.94 $\pm$ 0.07	0.82 $\pm$ 0.09
F1-score _{ADHD} (0–1)	0.88 $\pm$ 0.03	0.9 $\pm$ 0.02	0.88 $\pm$ 0.03	0.86 $\pm$ 0.05	0.95 $\pm$ 0.03	0.88 $\pm$ 0.03
F1-score _{Control} (0–1)	0.83 $\pm$ 0.04	0.85 $\pm$ 0.06	0.83 $\pm$ 0.06	0.78 $\pm$ 0.08	0.93 $\pm$ 0.04	0.83 $\pm$ 0.03
AUC (0–1)	0.86 $\pm$ 0.04	0.88 $\pm$ 0.03	0.86 $\pm$ 0.04	0.82 $\pm$ 0.06	0.94 $\pm$ 0.04	0.86 $\pm$ 0.03

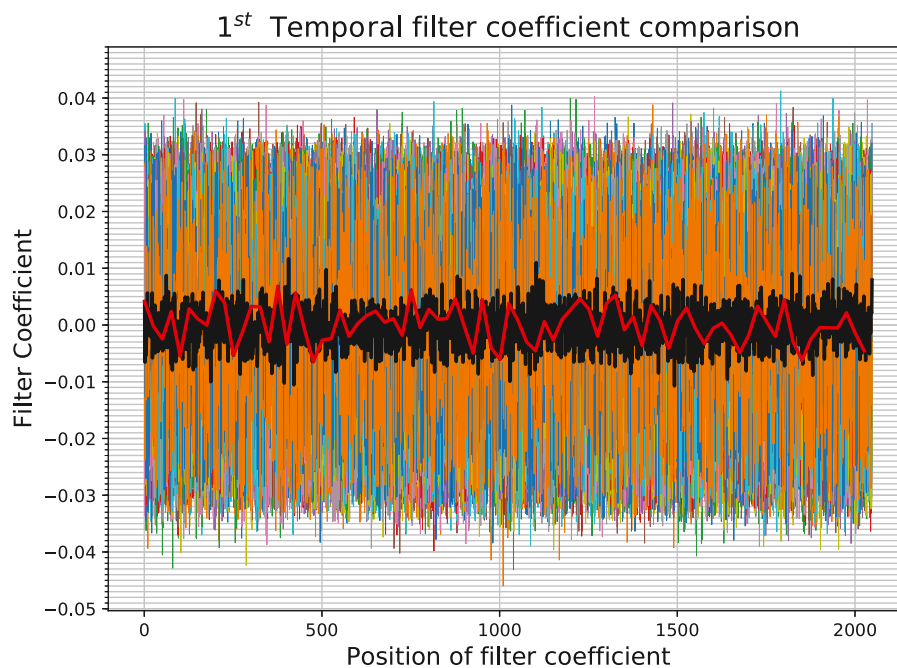


Fig. 7. Time response of the first temporal block filters in respect to the samples (the samples are in stacked version of tensors). the black line is the average response of the filters and because of rapid fluctuations, the red line with down-sampling scale equal to 50 is figured.

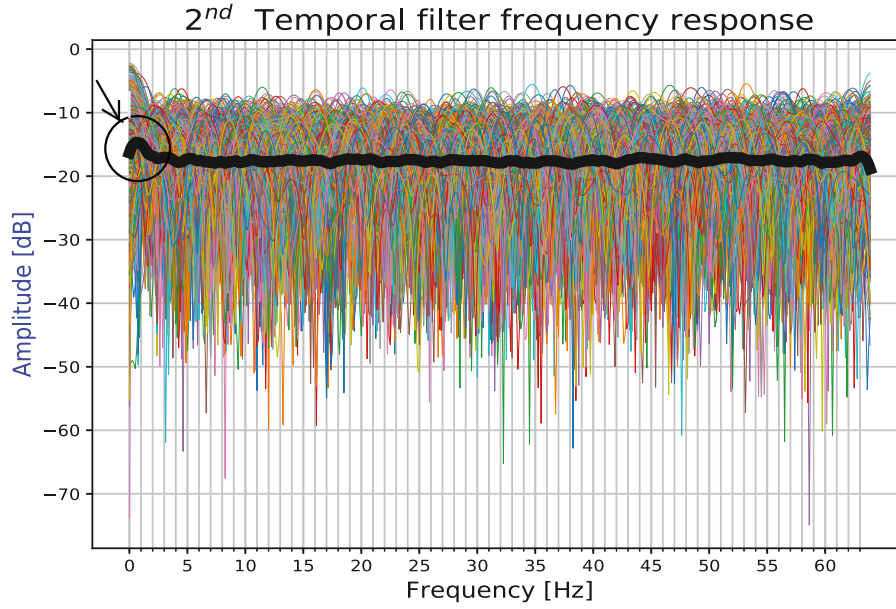


Fig. 8. Frequency response (in dB) of the second temporal block filters in respect to the Hz. The circled section shows that attention of the filters is on the delta band of EEG signals.

Table 4

Hybrid classification comparison on test set data using Flatten layer features maps (160-dimension).

Metric (Range)	NLSVM	LR	RF	GNB	KNN	CNN
Accuracy (0%–100%)	87.5	87.5	87.5	86.11	79.17	76.16
Precision _{ADHD} (0–1)	0.97	0.89	0.97	0.91	0.87	0.82
Precision _{Control} (0–1)	0.81	0.86	0.81	0.82	0.74	0.7
Recall _{ADHD} (0–1)	0.78	0.86	0.78	0.81	0.7	0.78
Recall _{Control} (0–1)	0.97	0.89	0.97	0.91	0.89	0.73
F1-score _{ADHD} (0–1)	0.87	0.88	0.87	0.86	0.78	0.8
F1-score _{Control} (0–1)	0.88	0.87	0.88	0.86	0.81	0.71
AUC (0–1)	0.88	0.88	0.88	0.86	0.79	0.76

position of the filters is 5th, 6th, and seventh. In Fig. 5 the coefficients are fluctuating around zero; however, the size of filters are $4 \times 1 \times 16$, and we stacked different dimensions together to build the 64×1 for making the visualization more effective. These two figures demonstrate that the spatial filters are focusing more on the middle of the electrodes in each convolution.

In the next figures, we investigated the temporal filters in both time and frequency analysis. Temporal filters in our proposed structure here would behave as FIR filters (because of causality and also the finite length of the filters), and this fact allows us to probe the frequency response of these filters. In Fig. 6, the frequency response (in dB) of each temporal filter (32 filters in shape of 1×128) and also the average of the responses (thick black) are shown. It is perfect for focusing on the circled section of the average response, which shows the attention of the extracted temporal filters on the alpha band, which has previously been demonstrated as a marker of ADHD in children [37]. In the other figure (Fig. 8) for the second temporal block, the circled section shows the attention of the filters on the delta band, which is another EEG marker for visual tasks in ADHD detection [38]. For better visualization of the temporal filters, the stacked version of both first and second blocks are shown in Fig. 7, and 9 respectively.

3.2. Results of combining different CNN layers outputs with classical classifiers

For measuring the performance of CNN and hybrid techniques with different classifiers, we applied other metrics such as accuracy, F1-score, each class recall, and precision (contains sensitivity, specificity). If we assume the confusion matrix, as mentioned in shape is illustrated in Fig. 10, we define all these metrics as Eq. (1):

$$\text{Accuracy} = \frac{TP + TN}{TN + FP + FN + TP}$$

$$\text{Precision}_{ADHD} = \frac{TP}{TP + FP}$$

$$\text{Precision}_{Control} = \frac{TN}{FN + TN}$$

$$\text{Sensitivity} = \text{Recall}_{ADHD} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \text{Recall}_{Control} = \frac{TN}{TN + FP}$$

$$\text{F1-score}_{ADHD} = \frac{2 * \text{Recall}_{ADHD} * \text{Precision}_{ADHD}}{\text{Recall}_{ADHD} + \text{Precision}_{ADHD}}$$

$$\text{F1-score}_{Control} = \frac{2 * \text{Recall}_{Control} * \text{Precision}_{Control}}{\text{Recall}_{Control} + \text{Precision}_{Control}}$$

AUC = Area under Receiver operating characteristic (ROC) curve using Pointwise Confidence Bounds [39]

The CNN can classify ADHD vs. non-ADHD epochs of EEG signals without any other classifiers. The performance of classification using the last layer's single neuron is shown in Table 2 for ten time repetition of model training, and then evaluating on the model using test dataset.

Since the CNN can automatically extract valuable features and extract them as the feature maps, it is helpful to check if it is possible to use its information in a hybrid format with other classical classifiers such as Non-linear SVM, LR, RF, GNB, KNN. In the following sections, we use feature maps from the output of the final three dense layers of CNN to train classical models and check these classifiers' performance. The classifiers' parameters are set consistent during different features from each layer.

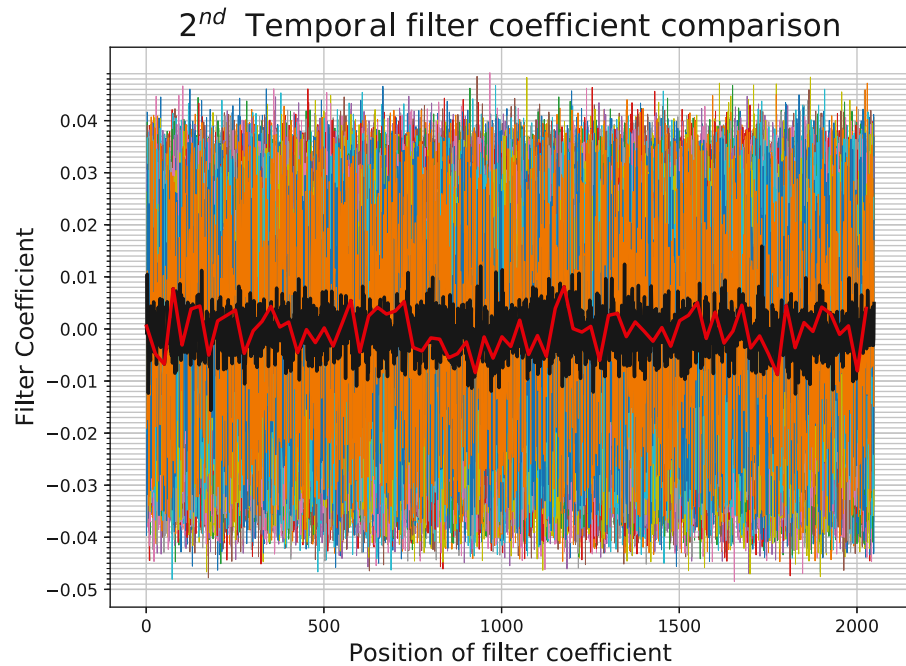
3.2.1. Combining feature maps from Flatten layer with classifiers

As mentioned in Table 1 and Fig. 3, the output of the flatten layer is 160-dimensional features that are passed through two spatial and two

Table 5

Results of hybrid classification scheme comparison using Dense 1 layer feature maps (64-dimension) in 5-fold cross-validation.

Metric (Range)	NLSVM	LR	RF	GNB	KNN	CNN
Accuracy (0%–100%)	87.20 ± 2.95	91.16 ± 0.03	85.68 ± 5.6	85.05 ± 4.19	93.9 ± 2.56	86.33 ± 2.64
Precision _{ADHD} (0–1)	0.92 ± 0.03	0.95 ± 0.04	0.9 ± 0.4	0.88 ± 0.07	0.95 ± 0.02	0.89 ± 0.04
Precision _{Control} (0–1)	0.82 ± 0.03	0.86 ± 0.02	0.8 ± 0.1	0.81 ± 0.03	0.93 ± 0.03	0.85 ± 0.07
Recall _{ADHD} (0–1)	0.86 ± 0.04	0.9 ± 0.02	0.85 ± 0.08	0.87 ± 0.01	0.95 ± 0.02	0.89 ± 0.07
Recall _{Control} (0–1)	0.89 ± 0.04	0.93 ± 0.05	0.87 ± 0.08	0.82 ± 0.09	0.92 ± 0.03	0.82 ± 0.09
F1-score _{ADHD} (0–1)	0.89 ± 0.03	0.92 ± 0.03	0.88 ± 0.05	0.87 ± 0.04	0.95 ± 0.02	0.88 ± 0.03
F1-score _{Control} (0–1)	0.85 ± 0.03	0.89 ± 0.03	0.83 ± 0.06	0.81 ± 0.05	0.92 ± 0.03	0.83 ± 0.03
AUC (0–1)	0.87 ± 0.03	0.91 ± 0.03	0.86 ± 0.05	0.85 ± 0.05	0.94 ± 0.03	0.86 ± 0.03

**Fig. 9.** Time response of the second temporal block filters in respect to the samples (the samples are in stacked version of tensors). the black line is the average response of the filters and because of rapid fluctuations, the red line with down-sampling scale equal to 50 is figured.**Table 6**Results of p-values^a for hybrid classification schemes in comparison to the CNN classification scheme using Dense 1 layer feature maps (64-dimension) in 5-fold cross-validation.

Metric	NLSVM	LR	RF	GNB	KNN
Accuracy	0.60	0.01	0.78	0.51	0.00 ^b
Precision _{ADHD}	0.19	0.02	0.43	0.79	0.02
Precision _{Control}	0.39	0.77	0.34	0.32	0.06
Recall _{ADHD}	0.45	0.83	0.40	0.63	0.11
Recall _{Control}	0.17	0.03	0.28	1.00	0.03
F1-score _{ADHD}	0.84	0.03	0.67	0.56	0.00
F1-score _{Control}	0.31	0.01	0.99	0.53	0.00
AUC	0.31	0.00	0.78	0.69	0.00

^aP-values less than 0.05 corresponds to significant increase in classification performance.^b0.00 means negligible value.

temporal blocks and focused more on discriminative details of the EEG signals. Since we had used ten repetitions for training the best model of CNN, we had ten performances for each classifier at this stage.

The overall performance of 5-fold cross-validation on the training set data is shown in Table 3. The results for each model (classifier) on the test data are shown in Table 4. With more in-depth analysis, we can

Table 7

Hybrid classification comparison on test set data using Dense 1 layer features maps (64-dimension).

Metric (Range)	NLSVM	LR	RF	GNB	KNN	CNN
Accuracy (0%–100%)	88.89	95.83	88.89	87.5	83.33	76.16
Precision _{ADHD} (0–1)	1	1	1	0.91	0.9	0.82
Precision _{Control} (0–1)	0.81	0.92	0.81	0.84	0.78	0.7
Recall _{ADHD} (0–1)	0.78	0.92	0.78	0.84	0.76	0.78
Recall _{Control} (0–1)	1	1	1	0.91	0.91	0.73
F1-score _{ADHD} (0–1)	0.88	0.96	0.88	0.87	0.82	0.8
F1-score _{Control} (0–1)	0.9	0.96	0.9	0.88	0.84	0.71
AUC (0–1)	0.89	0.96	0.89	0.88	0.84	0.76

conclude that the CNN, which is just a single neuron at the last layer performing the classification, can provide excellent results compared to the more complex classifiers. However, the KNN classifier works perfectly with these feature sets (160 dimensional) with the highest accuracy (and other metrics) among other classifiers.

After evaluating the models on the training set using a 5-fold cross-validation scheme, it is time to perform the final classification on the test (unseen) epochs. The results are shown in Table 4. Using this kind of feature (Flatten layer), the maximum accuracy on different classifiers

Table 8

Results of hybrid classification scheme comparison using Dense 2 layer feature maps (32-dimension) in 5-fold cross validation.

Metric (Range)	NLSVM	LR	RF	GNB	KNN	CNN
Accuracy (0%–100%)	85.36 $\pm$ 2.49	85.67 $\pm$ 4.04	84.77 $\pm$ 2.45	84.43 $\pm$ 3.37	88.1 $\pm$ 4.77	86.33 $\pm$ 2.64
Precision _{ADHD} (0–1)	0.89 $\pm$ 0.03	0.9 $\pm$ 0.05	0.89 $\pm$ 0.07	0.87 $\pm$ 0.03	0.91 $\pm$ 0.06	0.89 $\pm$ 0.04
Precision _{Control} (0–1)	0.8 $\pm$ 0.02	0.8 $\pm$ 0.06	0.79 $\pm$ 0.08	0.8 $\pm$ 0.06	0.84 $\pm$ 0.04	0.85 $\pm$ 0.07
Recall _{ADHD} (0–1)	0.86 $\pm$ 0.02	0.86 $\pm$ 0.05	0.85 $\pm$ 0.05	0.87 $\pm$ 0.04	0.88 $\pm$ 0.05	0.89 $\pm$ 0.07
Recall _{Control} (0–1)	0.84 $\pm$ 0.05	0.86 $\pm$ 0.06	0.86 $\pm$ 0.1	0.81 $\pm$ 0.06	0.88 $\pm$ 0.05	0.82 $\pm$ 0.09
F1-score _{ADHD} (0–1)	0.88 $\pm$ 0.02	0.88 $\pm$ 0.04	0.87 $\pm$ 0.02	0.87 $\pm$ 0.03	0.89 $\pm$ 0.05	0.88 $\pm$ 0.03
F1-score _{Control} (0–1)	0.82 $\pm$ 0.04	0.83 $\pm$ 0.04	0.81 $\pm$ 0.05	0.81 $\pm$ 0.05	0.86 $\pm$ 0.04	0.83 $\pm$ 0.03
AUC (0–1)	0.85 $\pm$ 0.03	0.86 $\pm$ 0.04	0.85 $\pm$ 0.04	0.84 $\pm$ 0.04	0.88 $\pm$ 0.05	0.86 $\pm$ 0.03

Confusion Matrix

Predicted \ Actual	Control (0)	ADHD (1)
Control (0)	True Negative (TN)	False Positive (FP)
ADHD (1)	False Negative (FN)	True Positive (TP)

Fig. 10. Confusion matrix definition which is specifically edited for our problem.**Table 9**

Hybrid classification comparison on test set data using Dense 2 layer features maps (32-dimension).

Metric (Range)	NLSVM	LR	RF	GNB	KNN	CNN
Accuracy (0%–100%)	83.33	90.28	84.72	83.33	93.06	76.16
Precision _{ADHD} (0–1)	0.88	0.94	0.91	0.86	0.92	0.82
Precision _{Control} (0–1)	0.79	0.87	0.8	0.81	0.94	0.7
Recall _{ADHD} (0–1)	0.78	0.86	0.78	0.81	0.95	0.78
Recall _{Control} (0–1)	0.89	0.94	0.91	0.86	0.91	0.73
F1-score _{ADHD} (0–1)	0.83	0.9	0.84	0.83	0.93	0.8
F1-score _{Control} (0–1)	0.84	0.9	0.85	0.83	0.93	0.71
AUC (0–1)	0.83	0.9	0.85	0.83	0.93	0.76

is 87.5%, and the highest performance in other metrics overall is related to random forest (RF) classifier.

3.2.2. Combining outputs of Dense 1 layer with classifiers

As we go deeper into the CNN structure, we can gain more essential details and features than the initial layers. Similar to the previous section, at this part, we use the feature maps of the Dense 1 layer (the notation is selected due to Fig. 3 diagram), which is a 64-dimensional feature set corresponding to a deeper sight and more optimized features chosen in comparison to the previous flatten layer features.

Table 7 shows each hybrid classifier method's performance to separate ADHD, Control EEG epochs from just 64 automatically extracted features. Also, for better insight into different classifiers, Table 5 shows the performance of each classifier using a 5-fold cross-validation scheme on training set utilizing Dense 1 output as the feature set.

As we had predicted before, this layer plays a feature selection function. It increases the separability of the two classes. The results also depict that the application of LR and KNN classifiers in combination with Dense 1 layer feature maps leads to drastically higher performance in comparison to the other classifiers. For accurate comparison and to evaluate the significant difference between different methods, the statistical t-test was performed between the results of only CNN structure and each hybrid model, and the results are shown in Table 6. As the table suggests, the LR and KNN classifiers have an overall more

significant increase in metrics in comparison to the CNN structure. This suggests that the LR and KNN may have more contribution in the unseen epochs.

The results of other hybrid classifiers are shown in Table 7, and it is shown that the LR classifier achieved high accuracy of 95.83% and the highest value in other metrics in comparison to the other hybrid methods. These results also show that our proposed CNN works well to extract separate features and focus on the raw EEG signals' most discriminating perspective details (see Table 8).

3.2.3. Combining outputs of Dense 2 layer with classifiers

We also used the output (features) from the third dense layer in our proposed CNN structure for the final evaluation. The features are 32-dimensional and better selected than the previous layers. But this feature reduction (selection) may lead to loss of information in some dimensions. However, the hypothesis about testing all of the classifiers with this feature set is made. The results of each hybrid method on 5-fold cross-validation on training epochs are shown in Table 9. As it is illustrated, the performance of these methods is relatively lower than the previous section (with 64-dimension features), mostly because of information loss and dimension composition due to the dimension reduction.

Furthermore, The detailed results on the test (unseen) set and epochs for each hybrid method are shown in Table 9. It is shown that this level of feature map extraction leads to lower performance in comparison to the previous dense layer feature maps. And it can be inferred that the performance is drastically affected by information reduction during feature selection, which is not optimal. However, at this stage of automatic feature extraction, again, the LR classifier could perform with 90.28% accuracy and almost the highest performance in other metrics such as Precision, and F1-score.

3.3. Comparing the results with the literature

Since there is no relevant literature that uses the deep neural networks on the raw EEG, we compared our best result related to 64-dimensional features maps from Dense 1 layer with three methods. The first article presents related work on manual feature extraction, which combines filter banks and time windowing methods in addition to nonlinear analysis of the EEG by extracting Fractal dimensions and Common Spatial Pattern (CSP) filters. As the work had been implemented on the same dataset and reached high accuracy compared to the previous literature, it is worthwhile to compare our results with that research. To have a better insight about other deep learning-based research, we candidate two papers with deep learning applications in children's ADHD detection using EEG. The first framework is related to the Chen et al. [31], which used Power Spectral Density (PSD) as their CNN's input with a reshaping scheme, and the second method is introduced by Dubreuil-Vall et al. which is published recently in 2020, and uses the spectrogram of EEG signals [33]. The results of our proposed method, which uses LR hybrid classifier with Dense 1 layer feature maps, are compared with these three schemes in our dataset, and the results are shown in Table 10.

Table 10

Comparison of the proposed method with literature (reported values are the values on test (unseen) epochs).

Metric (Range)	Proposed method (LR classifier+ Dense 1 features)	Manual feature extraction [26]	PSD + CNN [31]	ERSP + CNN [33]
Accuracy (0%–100%)	95.83	83.33	88.89	91.46
Precision _{ADHD} (0–1)	1.0	0.84	0.91	0.97
Precision _{Control} (0–1)	0.92	0.83	0.87	0.87
Recall _{ADHD} (0–1)	0.92	0.84	0.87	0.86
Recall _{Control} (0–1)	1	0.83	0.91	0.97
F1-score _{ADHD} (0–1)	0.96	0.84	0.89	0.91
F1-score _{Control} (0–1)	0.96	0.83	0.87	0.87
AUC (0–1)	0.96	0.83	0.89	0.92

As it is evident, the proposed method for classifying the ADHD, Control EEG signals without any other transformation, performed considerably better than the other methods which extract features manually or consider the information of the CNN with the time–frequency transformations on the main signals. Also, [33] performed better in ADHD Precision and Control Recall, which are compensated by our proposed method in other promising metrics such as accuracy, AUC, and sensitivity.

4. Conclusions

In this study, we proposed a new application of deep convolutional neural networks in ADHD disorder detection in children. We used the raw EEG signals without any transformations to feed the CNN structure. Also, there is no need for further artifact rejection methods, such as blind source separation (BSS), tensor decomposition, etc. We trained the CNN with two schemes to avoid over-fitting, and we nominated the best validation accuracy model selection based on higher performance compared to early stopping. Different classical classifiers were used to be coupled with our CNN neural network feature maps, and three additional feature maps from dense-shaped layers were extracted and evaluated to finalize the proposed framework. This evaluation investigated that using the feature maps from Dense 1 layer CNN and hybridizing with a logistic regression classifier leads to an average of 95.83% accuracy and 0.96 AUC. Compared to the previous literature from both manually and automatically (using deep neural networks) feature extraction schemes, these results demonstrate that our method performs relatively well in the case of ADHD detection in children using EEG signals. Our framework with this structure can be used in recognizing other types of mental disorders from EEG signals, while the concentration of the CNN is to extracting discriminative details of the signals and getting deeper the features become more accurate and optimally selected (if the information loss in dimension reduction does not happen).

Although our method can be well optimized to set up the number of filters and the size of kernels as the hyper-parameters, this scheme is reliable if a relatively high number of epochs in both categories restrict the research lack of adequate data situations.

Also, our framework suffers from a time-consuming training procedure, and for training and repeating the training procedure to increase the robustness of the model, more than 2 h on the system mentioned in this paper is required. Besides, our proposed method, like other deep learning-based methods, is highly data dependent. In other words, if the (condition of) data fed to the network change, the results of the network may alter.

Finally, this method focuses only on raw time series in both spatial and temporal domains. It means the overall training and feature extraction algorithms are based on the morphology of the signals, so any interruption and loss or unintended artifacts on signals in data acquisitions lead to wrong output. So, before any implementation, the data should be inspected, and this kind of trial should be removed or cleaned and then fed to the CNN.

CRediT authorship contribution statement

Behrad TaghiBeyglou: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing. **Ashkan Shahbazi:** Writing – original draft, Writing – review & editing. **Fatemeh Bagheri:** Writing – review & editing. **Sina Akbarian:** Methodology, Writing – review & editing. **Mehran Jahed:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Availability of data and material

The data which was used in this research is publicly available on the following website:

- [EEG data for ADHD / Control children \(Nasrabadi et al., 0000\)](#)
- [First National EEG Data Analysis Competition with Clinical Application \(NBML, 2019\)](#)

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